2 **(a)** Taking upwards is positive,

$$v_y^{\,2} = u_y^{\,2} + 2a_y s_y$$

$$0 = u_y^{\,2} + 2(-9.81)(1.8)$$

$$u_y = 5.93 \text{ m s}^{-1}$$

(b) Taking rightwards as positive,

$$F = \frac{(mv - mu)}{t}$$
$$= \frac{0.45(-5.2 - 6.6)}{0.22}$$
$$= -24.1\text{N}$$

Direction of the force on the ball is towards the left.

**marks were lost when students neglected the vector nature of the problem and subtracted the magnitudes of the two horizontal velocities.*

(c) Not elastic. The speed of the ball *v* has decreased from 6.6 m s^{-1} to 5.2 m s^{-1}. Thus, its kinetic energy $E_k = \frac{1}{2}mv^2$ has decreased.

(d) In this situation, air resistance is negligible and so the time for the ball to rise is equal to that for it to fall. However, the ball's horizontal velocity has decreased after the rebound. In the same time interval that it took to rise, the ball will thus travel a shorter horizontal distance after the rebound, and land closer to the wall.

3 **(a)** The mass is not uniformly distributed throughout the boat. The mass of the boat is